C5 Spark notebooks in Fabric

Differences from Databricks, environments, V-Order and Delta interoperability.

C5-L1 From Databricks notebooks to Fabric notebooks

Replace dbutils with notebookutils, secrets with Key Vault, inline installs with environments, and pass real timeouts to child notebooks.

Fabric replacement for dbutils	notebookutils (formerly mssparkutils).
Secrets in Fabric notebooks	notebookutils.credentials.getSecret(vault_url, name), from Azure Key Vault.
notebookutils.notebook.run default timeout	90 seconds. Pass timeout_seconds.
%pip in pipeline runs	Turned off by default. Use an environment.

C5-L2 Environments, resource profiles and V-Order

Pin libraries in a Full mode environment, keep the writeHeavy default for ingestion, and turn V-Order on for read-heavy gold tables.

Environment publishing modes	Quick: installs at session start (adds minutes). Full: reproducible snapshot, recommended for pipelines.
V-Order in new workspaces	Off by default (writeHeavy profile). Write cost around 15%, faster reads.
Turn V-Order on for one table	TBLPROPERTIES("delta.parquet.vorder.enabled" = "true").
spark.sql.parquet.vorder.enable	Removed in Runtime 1.3 and later. Use spark.sql.parquet.vorder.default or table and writer options.

C5-L3 Delta tables shared between Databricks and Fabric

Open format, but not every feature for every reader. Design shared gold tables for the weakest reader that needs them.

Delta interoperability rule	Every reader must support every feature the writer put in the table.
TIMESTAMP_NTZ in Fabric	Spark yes. SQL analytics endpoint and Direct Lake no.
Runtime 2.0 default table protocol	Reader 3, writer 7, deletion vectors on. Python notebooks cannot read deletion vectors.
Check a table's features	DESCRIBE DETAIL: minReaderVersion, minWriterVersion, table features.